Rank augmentation from a supported transposition

Human Compression project

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Abstract

We study a string marking (t, a, b, c) in which t is a single transposition and every orbit of $\langle a, b\rangle$ has at most three points. The normal closure of t gives the relevant end group explicitly. A dihedral stabilizer argument proves that both ends have the intersection property. Consequently the remaining obstruction is a small supported kernel. This proves the four- and five-point cases of the rank augmentation theorem of De Saedeleer–Leemans–Mulpas with their stated assumptions. For the six-point case it gives an exact test involving seven subgroup memberships and a useful sufficient orbit condition, replacing the false unrestricted assertion. All conclusions here are written proofs; a finite example is not used to establish a quantified statement.

1 The common setting

Let Ω be finite and let t, a, b, c be distinct nonidentity involutions in $\mathrm{Sym}(\Omega)$. Suppose that

$$t = (\xi\,\eta), \qquad b(\xi) = \xi,\ b(\eta) = \eta,\ c(\xi) = \xi,\ c(\eta) = \eta, \qquad ac = ca,$$

and that $G = \langle t, a, b, c\rangle$ is transitive. Put

$$B = \langle a, b\rangle, \qquad H = \langle t, a, b\rangle, \qquad K = \langle a, b, c\rangle.$$

Assume that every B-orbit has size at most three. Let O be the H-orbit containing ξ, η. We first show that its size is four, five or six; this is a consequence, not an additional assumption. All permutation products below act on the right.

The components for the two involutions a, b are singletons, one edge, a double edge, or a three-vertex path with successive labels a, b. At a vertex fixed by b, its component is a singleton, an a-edge, or the end of such a path. Each a-neighbor of an endpoint of t is fixed by c, because a commutes with c. The orbit O is precisely the union of the B-components containing the two endpoints. If neither component were a triple, c would fix this union pointwise and O would be a G-orbit. Transitivity would give $O = \Omega$ and $c = 1$, a contradiction. Thus at least one component is a triple. Such a component has only one point fixed by b, so the two endpoints of t lie in distinct B-components. Their union has size four, five or six. The marked paths on O, after a possible reversal, are exactly

$$\begin{array}{lcl} 4 & : & x -_t u -_a v -_b w, \\ 5 & : & x -_a y -_t u -_a v -_b w, \\ 6 & : & \ell -_b k -_a y -_t u -_a v -_b w. \end{array} \tag{1}$$

The letters in (1) are local point names. In each row, $T = \{u, v, w\}$ is a triple with u fixed by b, c and v fixed by c. In the six-point row the left triple has the same property with its order reversed.

We use the classical end criterion for string C-groups: two C-marked ends whose intersection is their common marked subgroup give a C-marking of the whole group. This is McMullen–Schulte, Proposition 2E16(a), quoted in [DLM, Proposition 5.1]. A pair of distinct involutions is a rank-two C-marking. We retain this criterion, rather than presenting it as new.

2 The two ends

Every three-point component gives the same marked quotient S_3 of B. On a two-point component, ab has order at most two. Consequently ab has order three or six, and B is dihedral of order six or twelve. Write

$$z = (ab)^3.$$

Then z is either the identity or a central involution of B; it fixes every triple pointwise. The kernel of the action of B on any one triple is $\langle z \rangle$. In particular, the stabilizers in B of u and w are $\langle b, z \rangle$ and $\langle a, z \rangle$ respectively.

Lemma 1. *Let $D = \langle p, q \rangle$ be generated by two involutions. Suppose q fixes v, while p sends v to $w \neq v$ and q does not fix w. An element of D fixing both v, w belongs to its cyclic rotation subgroup.*

Proof. Put $r = qp$, so $v^r = w$. Since q fixes v and normalizes $\langle r \rangle$, the D-orbit of v is its $\langle r \rangle$-orbit. This orbit has at least three points: it contains v, w, w^q, and w^q can equal neither w nor v (the latter equality would contradict $v^q = v$). Every element outside $\langle r \rangle$ conjugates r to r^{-1}. If such an element d fixed both v, w, then

$$v^r = w = w^d = (v^r)^d = (v^d)^{r^{-1}} = v^{r^{-1}}.$$

It would follow that $v^{r^2} = v$, contrary to the orbit size. □

Lemma 2. *Under the common hypotheses, both (a, b, c) and (t, a, b) are string C-markings.*

Proof. First consider $B \cap \langle b, c \rangle$. Every element of $\langle b, c \rangle$ fixes u, so the intersection lies in $\langle b, z \rangle$. If it were larger than $\langle b \rangle$, it would contain a nonidentity z. In the five-point row, b, c fix the left pair $\{x, y\}$ pointwise, whereas $z = (ab)^3$ interchanges it. This already gives a contradiction.

For the four- and six-point rows, choose an anchored triple whose terminal point is moved by c. Such a triple exists: otherwise c fixes every point of O, making O a G-orbit and forcing $c = 1$. Apply Lemma 1 to $p = b, q = c$ and the last two points v, w of that triple. Since z fixes both, it is a rotation in $\langle b, c \rangle$. It commutes with b, since it is central in B, and hence also with c. It already commutes with a, b. Moreover z fixes the endpoints of t in these two rows, so it commutes with t as well. A central permutation of a transitive group that fixes one point fixes every point: if $h(v) = v$, then $h(g(v)) = g(h(v)) = g(v)$. Thus $z = 1$, again a contradiction. We have proved $B \cap \langle b, c \rangle = \langle b \rangle$.

Next consider $B \cap \langle t, a \rangle$. This subgroup fixes w, so it lies in $\langle a, z \rangle$. A larger intersection than $\langle a \rangle$ would again put nonidentity z in the intersection. Apply Lemma 1 with $p = a, q = t$ at v, u. The t, a-orbit containing them has size three in the four-point row and size four in the other two rows. Hence z is a rotation in $\langle t, a \rangle$. On this orbit the rotation is a full three-cycle or four-cycle. Since z fixes u, its exponent is respectively divisible by three or four. Outside this orbit, t fixes every point, and the rotation acts as a, of order at most two. In the five- and six-point rows, divisibility by four therefore gives $z = 1$ globally. In the four-point row an even exponent also gives $z = 1$; the only other possibility is the restriction a_{out} of a to the complement of the three-point t, a-orbit.

In this last possibility, $a_{\text{out}} = a(u\,v)$. Since c fixes u, v and commutes with a, it commutes with a_{out}. The permutation $a_{\text{out}} = z$ commutes with a, b, and its support is disjoint from that of t. It is again central in G and has a fixed point, so it is the identity. Thus $B \cap \langle t, a \rangle = \langle a \rangle$. Both ends now satisfy the retained rank-three end criterion. □

3 The supported factor

If a permutation is defined on a subset of Ω, we extend it by the identity on the complement. Let B_{out} be the restriction image of B on $\Omega \setminus O$, viewed in this way as a permutation group.

Lemma 3. *In the four- and five-point rows,*

$$H = \mathrm{Sym}(O) \times B_{\mathrm{out}}. \tag{2}$$

In the six-point row let f be the unique bijection from the left triple to the right triple preserving the marked a, b actions. Let

$$N = \langle (v\ f(v)) : v \text{ in the left triple} \rangle \cong C_2^3.$$

Then

$$H = N \rtimes B. \tag{3}$$

These are identities of the actual permutation subgroups, not only abstract group isomorphisms.

Proof. Conjugating the pure transposition t by B gives transpositions on its orbit of unordered pairs. In the four-point row these pairs are the three edges of the star from x to T. They generate $\mathrm{Sym}(O)$. In the five-point row, the action of B on the left pair and the right triple is the full product $C_2 \times S_3$. Indeed z interchanges only the left pair on O, and the quotient action on T is S_3. The conjugate pairs are therefore all six edges of the complete bipartite graph between the pair and the triple. Again their transpositions generate $\mathrm{Sym}(O)$. In both cases these transpositions fix the complement of O pointwise. For any $g \in B$, multiply by the inverse of its restriction to O, already available in $\mathrm{Sym}(O)$, to obtain its pure complementary restriction. This proves (2), including independence of the two factors.

In the six-point row, B acts on its two triples through the same marked S_3 action, identified by f. The conjugates of t are exactly the three disjoint matching transpositions $(v\ f(v))$. They generate N, and B permutes these three generators. No nonidentity member of N preserves the two triples separately, whereas every member of B does. Thus $N \cap B = 1$ and (3) follows from $H = \langle t, B \rangle$. □

The remaining intersection can be expressed without testing all parabolic pairs. In (2), every $h \in H \cap K$ has the same complementary restriction as some $g \in B$. Hence hg^{-1} is supported on O. It follows that

$$H \cap K = B \quad \Longleftrightarrow \quad K \cap \mathrm{Sym}(O) = B \cap \mathrm{Sym}(O). \tag{4}$$

This is the elementary restriction-kernel test: the actual images are already known, so only their kernels remain to be compared. In (3), the unique factorization $h = ng$ gives instead

$$H \cap K = (N \cap K)B, \qquad H \cap K = B \quad \Longleftrightarrow \quad N \cap K = 1. \tag{5}$$

The product in (5) is a semidirect product, since B normalizes $N \cap K$.

4 Blocks and parity in the four- and five-point rows

Removing the single t-edge disconnects the small left component in (1), a singleton or a pair. In either row K fixes that component setwise. Every other point belongs to one K-orbit, say R: the full labelled graph is connected, and the removed edge was its sole edge at that small component. In particular every triple outside O lies in R.

Lemma 4. *Let R be any K-orbit containing an anchored triple T, and suppose K acts imprimitively on R. Every B-triple in R has its three points in distinct blocks of each nontrivial system of blocks. Moreover, there is another B-triple $T' \subseteq R \setminus T$ whose three points lie respectively in the same three blocks as T. In the four- and five-point rows, T' lies outside O.*

Proof. The equivalence relation induced on any B-triple by the blocks is invariant under its full S_3 action. It is either universal or discrete. Suppose a triple U lay in one block D. Then a, b preserve D. If c also preserved D, transitivity on R would give $D = R$. Otherwise put $D' = D^c \neq D$. These blocks are disjoint, c interchanges them, and a preserves D' because it commutes with c. The images under c

of the two a-adjacent points of U are a-adjacent points v', w' in D'. They cannot both be sent outside D' by b: their two distinct images, together with v', w', would belong to one B-orbit of at least four points. Thus b sends a point of D' into D', and hence preserves that whole block. The union $D \cup D'$ is now invariant under a, b, c, so equals R. But c interchanges its two blocks, contradicting the fixed point u of c in the anchored triple. This proves the first assertion.

All blocks have the same size, at least two. Choose $u' \neq u$ in the block of u. The points u', $v' = (u')^a$, and $w' = (u')^{ab}$ lie in the three distinct blocks of u, v, w. They belong to one B-orbit, which has at most three points. Its marked action is therefore the triple $u' -_a v' -_b w'$. This orbit is disjoint from T: if it met T the two B-orbits would coincide, whereas u' is not any point of T. In the four- and five-point rows the other points of O belong to the small left K-orbit, so this new triple is outside O. □

For comparison with [DLM, Lemma 5.3], every imprimitive K-orbit in our setting contains an anchored triple. Indeed, removing one edge from the connected labelled graph leaves at most two components, each containing an endpoint of that edge. If its B-component is a singleton or a pair, c fixes that whole component and its K-orbit has that same size, admitting no nontrivial block system. Otherwise the component is an anchored triple. Thus Lemma 4 proves both conclusions of the source lemma without its extended graph propagation argument.

Theorem 5. *In the common setting, the output* (t, a, b, c) *is a string C-marking if one of the following holds.*

1. $|O| = 4$*, and* K *acts imprimitively on its orbit containing* T *whenever that orbit contains another* B*-triple.*

2. $|O| = 5$*, the permutations* a, b, c *are even, and the same imprimitivity condition holds.*

Proof. By Lemma 2, it suffices to prove $H \cap K = B$. Suppose first that there is a triple outside O. The relevant K-orbit is R, so the assumed imprimitivity and Lemma 4 give a triple T' outside O sharing the three blocks of T. Take $h \in H \cap K$, and use (2) to choose $g \in B$ with the same action outside O. Then $h' = hg^{-1}$ fixes T' pointwise and lies in K; it consequently preserves all three specified blocks. It also preserves T setwise, since it preserves O and the small left K-orbit. Each of these blocks contains exactly one point of T, so h' fixes T pointwise. In the four-point row the remaining point of O is fixed by K, whence $h' = 1$. In the five-point row the only other possibility is the single transposition of the left pair. This is excluded by $K \leq \mathrm{Alt}(\Omega)$. Thus $h = g \in B$.

It remains to handle the case with no triple outside O, so no imprimitivity assumption is available. Let a_{out} be the restriction of a to $\Omega \setminus O$. On that complement all B-orbits have size at most two; hence a_{out} commutes with b. It commutes with a and t as well. The restriction of a to O has support fixed pointwise by c, so $ac = ca$ also gives $a_{\mathrm{out}}c = ca_{\mathrm{out}}$. Centrality in transitive G and its fixed points on O force $a_{\mathrm{out}} = 1$. Thus $B_{\mathrm{out}} = \langle b_{\mathrm{out}} \rangle$ has order at most two.

For $|O| = 4$, $B = \mathrm{Sym}(T) \times \langle b_{\mathrm{out}} \rangle$. Indeed, if $b_{\mathrm{out}} \neq 1$, then $(ab)^3 = b_{\mathrm{out}}$ isolates its complementary factor; the restrictions of a, b generate $\mathrm{Sym}(T)$. Every element of $H \cap K$ fixes the remaining point of O and has complementary action in $\langle b_{\mathrm{out}} \rangle$, so belongs to this very group B.

For $|O| = 5$, evenness of b forces $b_{\mathrm{out}} \neq 1$, since its restriction to O is a single transposition. The group

$$L = \mathrm{Sym}(\{x, y\}) \times \mathrm{Sym}(T) \times \langle b_{\mathrm{out}} \rangle$$

has order 24. Its even subgroup has order 12, and equals B: B has order 12 by its faithful $C_2 \times S_3$ action on O, and both a, b are even. Every member of $H \cap K$ preserves the left pair and T, has complementary action in $\langle b_{\mathrm{out}} \rangle$, and is even. It therefore belongs to B. This proves both cases, including the boundary where the source's imprimitivity condition is vacuous. □

5 An exact replacement for the six-point assertion

Theorem 6. *In the six-point row of (1), the marking (t, a, b, c) has the intersection property if and only if none of the seven nonidentity products of the three matching transpositions belongs to K. In particular it has that property if the left and right triples of O lie in distinct K-orbits.*

Proof. Both ends have the intersection property by Lemma 2. Equation (5) and the end criterion show that the whole marking has it exactly when $N \cap K = 1$. Since N is generated by three disjoint transpositions, its seven nonidentity elements are precisely the listed products. If the two triples lie in different K-orbits, each nonidentity member of N moves a point from one such orbit to the other and so cannot belong to K. □

The last condition is a genuine sufficient restriction, not a consequence of the source hypotheses. The separately frozen eighteen-point example has $K = S_{18}$ and contains t itself; it fails the first of the seven tests. Nor does the theorem above merely repeat the full intersection property: it replaces all parabolic intersections by membership in one explicit group of order eight, after separately proving the two end properties uniformly.

6 The source application and attribution

Start with the exact hypotheses of [DLM, Theorem 4.1]: a natural S_n rank-three C-marking (a, q, c), $n \geq 5$, with q not a transposition. Choose a transposition t in q whose endpoints are fixed by c, and put $b = tq$. Then b is an involution, fixes those endpoints, and commutes with t; also c commutes with t and a. The new marking generates S_n, since $q = tb$. The orbit argument in Section 1 proves, without selecting one of the three cases, that the orbit containing t has size four, five or six. The local paths (1) also show that the four marks are distinct and nonidentity. Thus all common hypotheses above are satisfied when every B-orbit has size at most three. Theorem 5 proves the source's cases (1) and (2) with their full original assumptions. Theorem 6 gives the stated corrected replacement for case (3), whose unrestricted assertion is refuted in the separate exact counterexample.

The rank augmentation operation, cases (1) and (2), local component analysis and intended end-intersection method belong to De Saedeleer, Leemans and Mulpas. Our normal-closure and restriction-kernel formulations reuse the elementary supported-transposition method from our common gluing proof. The short dihedral and block arguments here replace the longer local propagation arguments needed for these conclusions; we do not claim that the underlying group facts are historically new. The standard end criterion remains an attributed external premise. Two declarations in `SupportedIntersections.lean` prove the actual arbitrary-group factorization tests behind (4) and (5), using subgroups and their intersections. The actual permutation factorizations in Lemma 3, the end arguments, block and parity arguments, and source application are written proofs; they are not silently included in that formal certificate. The proof includes both conclusions of source Lemma 5.3. The companion application verifies the source's three Section 6 examples using its complete required program. The introductory Table 1 remains explicitly retained motivating data; it is not a newly regenerated census.

References

[1] J. De Saedeleer, D. Leemans and J. Mulpas, *A rank augmentation theorem for rank 3 string C-group representations of the symmetric groups*, arXiv:2212.12563v1, 23 December 2022, Theorem 4.1 and Section 5.